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Multiple choice questions generated below. Answers in RED and incorrect answers in BLUE.

Q1. Which of the following statements about physical idealizations is most reasonable?

A. Treating a horse as a perfect sphere is appropriate when calculating the bending of its knee joint while running.
Joint-level motion depends on the animal's shape and internal structure, so a sphere model throws away the key geometry.

B. Treating a horse as a point mass is appropriate when estimating how far it travels across a multi-kilometer field over several minutes.
Over distances far larger than the animal's size, the center-of-mass position change can be modeled well by a point mass.

C. It is invalid to model a situation as vacuum-like under any circumstances, even if drag forces are negligible.
"Ignoring air resistance" is precisely an idealization that approximates vacuum when drag is small compared with other forces.

D. None of the above is reasonable.
Option B is a standard and defensible approximation for large-scale translational motion.

Q2. Which of the following equations is correct?

A. 1 nickel = 1 $
A nickel is only a small fraction of a dollar, not equal to a full dollar.

B. 1 = 0.05
This drops the units and incorrectly asserts a pure number equality.

C. 1 nickel = 0.05 $
A nickel is defined as five cents, i.e., 0.05 dollars.

D. 20 nickels = 0.05 $
Twenty nickels is $20 \times 0.05 = 1.00$ dollar, not 0.05 dollars.

Q3. Which option has the incorrect unit?

A. Power: watt (W)
A watt is the correct SI unit of power (J/s).

B. Pressure: newton/meter² (N/m²)
Pressure is force per area, so $\mathrm{N/m^2}$ (pascal) is correct.

C. Electric charge: coulomb/second (C/s)
C/s is the unit of electric current (ampere), while charge alone is measured in coulombs (C).

D. Momentum: kilogram·meter/second (kg·m/s)
Momentum is mass times velocity, giving SI units of kg·m/s.

Q4. A scout leaves the fortress at the origin and spots an enemy camp at the position vector $\vec{r} = -5\,\hat{\imath} + 12\,\hat{\jmath}$, where distances are measured in kilofurlongs. Which statement is correct?

A. The camp is 17 kilofurlongs from the fortress.
17 would come from incorrectly adding magnitudes $(5 + 12)$ instead of using the Pythagorean theorem.

B. The camp is 12 kilofurlongs from the fortress.
12 is just the y-component, not the full distance from the origin.

C. The camp is 5 kilofurlongs from the fortress.
5 is just the magnitude of the x-component, not the total displacement length.

D. None of the above is correct.
The distance is $\sqrt{(-5)^2 + 12^2} = \sqrt{169} = 13$ kilofurlongs, which is not 17, 12, or 5.

Q5. If the displacement of an object is given by the function $\vec{s} = \left((0.90\,\mathrm{m/s^3})t^3 - (1.60\,\mathrm{m/s^2})t^2\right)\hat{\imath}$, which of the following statements is correct?

A. This motion has constant acceleration.
Because $s(t)$ contains a t^3 term, the acceleration ends up depending on time rather than being constant.

B. The magnitude of the acceleration is 1.60 m/s².
The coefficient in the t^2 term is not the acceleration; acceleration comes from the second time derivative of position.

C. The magnitude of the acceleration at $t = 0$ is 0 m/s².
At $t = 0$, the second derivative is generally nonzero when a t^2 term is present.

D. The magnitude of the acceleration at $t = 0$ is 3.20 m/s².
With $s(t) = 0.90t^3 - 1.60t^2$, $a(t) = d^2s/dt^2 = 5.40t - 3.20$, so $|a(0)| = 3.20$ m/s².

Q6. An object takes 5 s to move from position $\vec{r}_A = 1.0\,\hat{\imath} - 2.0\,\hat{\jmath}$ to position $\vec{r}_B = 7.0\,\hat{\imath} + 4.0\,\hat{\jmath}$ along a straight path, where length units are in meters. Which statement is correct?

A. The moving distance is 8.5 m.
The straight-line distance is $|\Delta\vec{r}| = \sqrt{(7-1)^2 + (4-(-2))^2} = \sqrt{36+36} = \sqrt{72} \approx 8.5$ m.

B. The speed at $t = 0$ must be 1.7 m/s.
Knowing only start/end points and total time does not determine the instantaneous speed at the start.

C. The acceleration at $t = 0$ must be 0.34 m/s².
No acceleration information is implied by just the endpoints and travel time along a straight path.

D. None of the above is correct.
Option A is correct because the straight-path distance is fixed by geometry.

Q7. A soccer ball is kicked with initial speed u_0 at an angle α above the horizontal. Neglecting air resistance, what is the ball's speed at its maximum height?

A. u_0.
At the top, the vertical component has dropped to zero, so the speed is less than the launch speed unless $\alpha = 0$.

B. $u_0 \cos \alpha$.
At maximum height the vertical velocity is zero while the horizontal component remains $u_0 \cos \alpha$, so the speed equals that value.

C. $u_0 \sin \alpha$.
This is the initial vertical component, which becomes zero at the highest point.

D. 0.
Only the vertical component becomes zero at the peak; the horizontal component generally remains nonzero.

Q8. A drone begins with initial velocity $\vec{v}_0 = (15.0\,\mathrm{m/s})\,\hat{\imath}$ and then experiences a constant acceleration $\vec{a} = (-1.20\,\mathrm{m/s^2})\,\hat{\imath} + (2.80\,\mathrm{m/s^2})\,\hat{\jmath}$. Which statement is correct?

A. The x-component of the drone's velocity remains constant.
Because $a_x \neq 0$, the x-component of velocity must change with time.

B. The y-component of the drone's velocity remains constant.
Because $a_y \neq 0$, the y-component of velocity must change with time.

C. Both the direction of motion and the speed change.
Nonzero acceleration components in both x and y change the velocity vector's components, altering both its magnitude (speed) and direction.

D. None of the above is correct.
Option C correctly follows from having acceleration in two perpendicular directions.

Q9. If the magnitude of each of two point charges is doubled and their separation distance is tripled, what happens to the magnitude of the electrostatic force between them?

A. It stays the same.
Scaling does not cancel here because the charges scale the numerator and the distance scales the denominator differently.

B. It becomes $\frac{4}{9}$ of the original.
By Coulomb's law $F \propto \frac{q_1 q_2}{r^2}$, doubling both charges gives a factor 4 and tripling distance gives a factor 1/9, so net factor is 4/9.

C. It increases by a factor of 2.
This common error accounts for changing one charge but ignores the second charge and the r^2 dependence.

D. It increases by a factor of 4.
This includes the charge scaling but incorrectly forgets that increasing separation reduces the force by r^2.

E. It decreases by a factor of 3.
Distance affects the force through r^2, so a factor of 3 in distance leads to a factor of 9 in the denominator, not 3.

Q10. A planet's escape speed is given by $v_{\text{esc}} = \sqrt{\frac{2GM}{R}}$. Based on this expression, which statement is correct?

A. A heavier spacecraft needs a larger escape speed than a lighter spacecraft on the same planet.
Escape speed depends on the planet's M and R, not on the mass of the escaping object.

B. If a planet's radius increases while its mass stays the same, the escape speed increases.
Since $v_{\text{esc}} \propto 1/\sqrt{R}$, increasing R actually decreases the escape speed when M is fixed.

C. If a planet's mass decreases while its radius also decreases, the escape speed must increase.
Changing M and R in opposite directions does not guarantee an increase because v_{esc} depends on the ratio M/R.

D. A planet with larger mass and smaller radius requires a higher escape speed.
Because $v_{\text{esc}} \propto \sqrt{M/R}$, increasing M and decreasing R both raise the escape speed.

Q11. A constant force $\vec{F} = (-2.50\,\text{N})\,\hat{\imath} + (6.00\,\text{N})\,\hat{\jmath}$ acts on a particle as it moves along the y axis from $y = -1.0\,\text{m}$ to $y = 3.0\,\text{m}$. What is the work done by the force on the particle?

A. 24 J.
The displacement is $\Delta\vec{r} = 4.0\,\hat{\jmath}$ so $W = \vec{F} \cdot \Delta\vec{r} = (6.00)(4.0) = 24$ J and the x-component does no work.

B. −10 J.
This comes from incorrectly using only the x-component of force even though the motion is purely along y.

C. 14 J.
This kind of value typically results from mixing components or using the wrong displacement magnitude.

D. 0 J.
Work is not zero because the force has a nonzero component along the displacement direction (the y-direction).

Q12. A 3.0 kg cart moves toward a bumper at 5.0 m/s, hits it, and rebounds straight back at 2.0 m/s. The contact time is 0.15 s. What is the magnitude of the average force on the cart during the collision?

A. 60 N.
This results from using $|v_f - v_i| = 3$ m/s instead of accounting for the velocity reversal.

B. 100 N.
This typically comes from dividing by the wrong collision time or omitting part of the momentum change.

C. 140 N.
Taking toward the bumper as positive gives $v_f = -2.0$ m/s, so $|F_{\text{avg}}| = \left|\dfrac{m(v_f - v_i)}{\Delta t}\right| = \left|\dfrac{3.0(-2.0 - 5.0)}{0.15}\right| = 140$ N.

D. None of the above is correct.
A correct impulse calculation yields 140 N, so an answer is listed.

Q13. A 70 kg catcher on frictionless ice is initially at rest. A 145 g baseball traveling at 65 mph is caught and sticks in the glove. What is the catcher's speed immediately after the catch?

A. 0.060 m/s.
Using momentum conservation for a perfectly inelastic catch, $v_f = \dfrac{mv}{M+m} = \dfrac{0.145 \times (65\,\text{mph} \approx 29.1\,\text{m/s})}{70.145} \approx 0.060\,\text{m/s}$.

B. 0.22 m/s.
This overestimates the result by not dividing by the combined mass (or by treating the catcher's mass as much smaller).

C. 2.1 m/s.
This mistake comes from confusing the catcher's final speed with the baseball's initial speed scale.

D. 13 m/s.
This error typically happens when someone forgets the huge mass difference and effectively ignores the catcher's mass.
